

SERAK, Czech Republic

WMO: 117300

Lat: 50.188N

Lon: 17.108E

Elev: 4355

StdP: 12.53

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.6	(c) 7.0	(d) -10.1	(e) 3.7	(f) 16.0	(g) -3.5	(h) 5.4	(i) 17.9	(j) 42.3	(k) 23.3	(l) 39.0	(m) 23.5	(n) 16.7	(o) 100	(p) 0.816

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.2	(c) 72.3	(d) 59.7	(e) 69.1	(f) 58.9	(g) 66.3	(h) 57.7	(i) 61.7	(j) 68.4	(k) 60.2	(l) 66.7	(m) 58.8	(n) 65.2	(o) 8.4	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	88.9	64.3	57.6	83.3	63.2	56.0	78.6	61.9	29.8	68.8	28.6	66.5	27.6	65.3	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
34.0	30.2	27.1		-1.7	76.9	7.2	2.4	-6.9	78.6	-11.1	80.0	-15.1	81.4	-20.4	83.1	
			DB	-2.3	64.9	7.1	1.5	-7.4	66.0	-11.5	66.9	-15.5	67.7	-20.6	68.8	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.8	22.3	22.5	27.7	38.2	45.0	51.9	55.7	55.6	47.4	39.8	32.4	25.6
	DBStd	14.29	7.52	8.60	8.58	8.38	7.98	7.57	6.95	6.72	7.21	7.90	8.22	7.43
	HDD50	4736	859	771	692	361	197	68	25	21	133	326	527	756
	HDD65	9599	1324	1191	1157	804	620	396	298	300	529	782	977	1221
	CDD50	633	0	0	0	7	42	125	200	195	55	9	0	0
	CDD65	21	0	0	0	0	0	3	9	9	0	0	0	0
	CDH74	32	0	0	0	0	0	2	17	12	1	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	13.4	17.1	15.6	14.5	12.0	11.5	10.4	10.2	10.1	12.0	14.3	16.0	16.9
Precipitation	PrecAvg	34.6	2.4	2.1	2.5	2.2	3.3	4.0	4.7	3.4	3.1	2.3	2.4	2.4
	PrecMax	47.8	5.0	4.4	5.9	3.5	8.5	7.2	19.9	7.8	7.1	6.5	3.9	4.5
	PrecMin	23.4	1.0	0.5	0.9	0.2	1.3	1.5	1.3	0.9	0.7	0.3	0.0	0.7
	PrecStd	4.9	1.1	1.0	1.1	0.9	1.5	1.4	3.5	1.6	1.8	1.4	0.9	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.9	44.0	49.2	61.3	68.3	73.4	76.4	75.5	66.6	61.2	54.0	43.9
		MCWB	34.3	31.8	39.1	49.5	56.1	61.5	61.4	59.8	57.3	48.6	39.8	33.5
	2%	DB	37.0	38.5	44.9	56.4	63.0	69.1	72.5	71.9	62.7	55.6	49.2	40.6
		MCWB	34.8	32.1	38.5	46.7	54.0	59.0	59.8	59.1	55.4	48.4	42.2	34.1
	5%	DB	34.8	34.8	41.9	52.8	59.7	65.8	69.0	68.7	60.0	52.5	45.7	38.2
		MCWB	32.5	32.3	36.6	45.1	52.0	57.8	58.9	58.8	53.4	47.6	42.1	34.2
	10%	DB	32.2	32.3	38.9	49.4	56.4	62.7	66.0	65.8	57.0	50.0	43.2	35.2
		MCWB	30.7	30.6	36.2	43.1	50.2	55.8	57.6	57.7	51.8	46.5	40.4	32.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	36.4	42.1	50.8	58.3	63.5	64.7	62.8	58.9	53.2	47.1	39.4
		MCDB	38.4	38.0	45.8	58.4	65.9	69.8	71.2	69.8	64.7	55.5	48.8	39.9
	2%	WB	35.7	34.4	40.0	47.8	55.1	61.1	61.7	61.2	56.5	50.3	44.9	37.6
		MCDB	36.1	35.6	43.0	54.8	61.3	66.6	68.9	67.8	61.3	52.9	46.4	38.6
	5%	WB	33.3	32.6	38.2	45.8	53.0	58.9	60.0	59.8	54.4	48.5	43.0	35.4
		MCDB	34.1	34.0	40.9	51.6	58.1	64.7	66.8	66.5	58.4	51.2	44.7	36.6
	10%	WB	31.1	30.8	36.2	44.0	50.9	56.6	58.5	58.5	52.7	46.9	41.0	32.9
		MCDB	31.6	32.5	38.9	48.8	55.8	61.7	64.8	65.1	56.2	49.5	42.4	33.9
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.4	8.5	10.3	10.8	10.9	11.2	11.2	9.0	8.0	7.3	7.1
		MCDBR	9.0	10.3	11.6	13.4	13.7	13.8	14.4	13.9	11.8	10.5	10.3	9.7
		MCWBR	7.6	8.0	7.9	7.8	8.1	8.6	8.0	7.5	7.9	7.4	7.1	7.2
	5% WB	MCDBR	8.1	9.2	10.3	12.4	12.9	13.1	13.3	12.8	11.0	9.2	8.4	8.8
		MCWBR	7.6	8.3	7.7	7.7	8.4	8.7	8.5	7.9	7.7	7.4	6.9	7.7
Clear-Sky Solar Irradiance	taub	0.262	0.283	0.315	0.354	0.352	0.352	0.364	0.358	0.332	0.312	0.291	0.261	
	taud	2.392	2.391	2.377	2.327	2.354	2.371	2.334	2.364	2.428	2.471	2.476	2.440	
	Ebn at Noon	253	275	284	282	287	287	281	277	274	260	239	238	
	Edh at Noon	19	25	31	36	37	37	38	35	29	23	18	16	
All-Sky Solar Radiation	RadAvg	265	501	838	1312	1588	1683	1602	1439	1017	593	291	200	
	RadStd	47	75	137	166	190	158	182	132	149	88	38	22	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		+0.90	N/A	N/A	+1.35	+0.63	+0.56	N/A	-274	+117	+49

Nomenclature: See separate page